

ALYSSA HWANG

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Human-AI Systems ◇ User-driven research and product direction

SUMMARY

PhD in computer science specializing in human-AI interaction. Experienced in evaluating emerging AI capabilities, understanding user needs, and building products that connect the two. Brings deep technical knowledge, strong communication skills, and a practical approach to helping organizations make better decisions about AI.

EDUCATION

University of Pennsylvania	Philadelphia, PA
<i>PhD</i> in Computer and Information Science	Dec 2025
<i>MS</i> in Computer and Information Science	Dec 2023
<i>Advisors:</i> Boon Thau Loo, Andrew Head	
<i>Thesis:</i> User Interfaces for Fine-grained Integration of Information	

Columbia University	New York, NY
<i>BS</i> in Computer Science	May 2020
<i>Advisor:</i> Kathleen McKeown	
<i>Thesis:</i> Towards Augmenting Lexical Resources for Slang and African American English	

SKILLS

Research: User-centered design, experimental design, usability studies, interviews, surveys, data analysis
Technical: Python, JavaScript, React, C/C++, web app development, LLM framework/API integration
Product: Product strategy, roadmap definition, stakeholder communication, cross-functional collaboration

FEATURED WORK

Sage Notebook: AI-Native College Advising Platform

Founder & Product Engineer ◇ Product strategy, full-stack development, LLM integration

- Created an AI-native college advising platform from concept to MVP, owning product strategy, UX design, and full-stack engineering across LLM integration, database architecture, authentication, deployment, and payments while collaborating with business and design partners.

AI-Augmented Reading Platform for Multimedia Documents

Lead Researcher ◇ Human-AI interaction, rapid software prototyping, user evaluation

- Identified user needs and prototyped an AI-augmented reading platform for multimedia documents; integrated LLM capabilities into an interactive web interface; evaluated usability through comparative studies.

Empirical Analysis of Voice Interaction Challenges and Opportunities

Lead Researcher ◇ User research, interaction design, problem framing, AI product strategy

- Studied real-world voice assistant interactions through in-home observational research with participants using Alexa, applying thematic analysis and grounded theory to uncover unmet needs, characterize voice interaction challenges, and identify opportunities for AI-powered experiences.

EXPERIENCE

University of Pennsylvania, PhD Researcher

Sept 2020–Dec 2025

Penn HCI & Penn NLP

- Led end-to-end research and development of AI-powered systems, combining NLP, HCI, and user-centered design to create tools for understanding complex information.
- Collaborated across research labs and disciplines, publishing at leading NLP and HCI venues while contributing to the growth of Penn HCI as a founding member.

Amazon Web Services, Applied Scientist Intern

May–Sept 2022

NewsQs: Multi-Source Question Generation for the Inquiring Mind ([arXiv preprint](#))

- Developed a cross-task data generation pipeline that transformed a multi-document summarization dataset into a multi-document question answering dataset using fine-tuned T5 models.
- Partnered with Machine Learning Data Linguists team to co-design a human-centered annotation framework.

Google, Software Engineering Intern

May–Aug 2021

Image Ranking for Shopping

- Established an image clustering algorithm to diversify shopping review photos and identify incorrect product images; contributed tools and documentation enabling non-technical users to interpret system outputs.

Google, Software Engineering Intern

June–Aug 2020

Dialogue Act Recognition for Dialogflow

- Trained a convolutional neural network to predict dialogue act labels for conversational AI applications.

Google, Software Engineering Intern

May–Aug 2019

Image Carousel for Events Search

- Implemented a backend pipeline to surface relevant images from internal APIs for Google Events Search.

LEADERSHIP & SERVICE

Penn CIS PhD Mentorship Program, Founder

May 2022–Dec 2025

- Founded and scaled a department-wide mentorship program by securing funding, building operational processes, and recruiting volunteers to organize recurring community events.
- Planned for long-term sustainability through leadership succession planning, transitioning ownership while maintaining participation from over two-thirds of CIS PhD students.

Computer and Information Science Doctoral Association, Co-Chair

Sept 2020–May 2023

- Elected as Chair-in-Training and later served as Co-Chair, facilitating communication between CIS PhD students, faculty, and department leadership.
- Received Outstanding Service Award from Penn Engineering for contributions to the CIS doctoral community.

MindCORE DivE In, Programming Development Committee Lead

Mar–Oct 2021

- Co-developed a funded outreach program supporting prospective graduate students in the mind sciences; led workshop programming to introduce research pathways and graduate education opportunities.

INVITED TALKS

Hop, Skim, and a Jump from Text to Voice

Apr 15, 2026

SAP Futures Research

Authorship Matters in Generative AI

Nov 20, 2024

The Ethics Institute at Kent Place School

A Principled Approach to Developing Grounded Intuition of Generative Models Apr 18, 2024
Google PAIR (People and AI Research)

Rewriting the Script: Human-Centered Opportunities for NLP Apr 17, 2024
Columbia University Natural Language Processing Seminar

Human-centered AI Research and Design Feb 9, 2024
Women in Data Science @ Penn

SELECTED PUBLICATIONS AND PREPRINTS

Connecting the Dots: Surfacing Structure in Documents through AI-Generated Cross-Modal Links
Alyssa Hwang, Hita Kambhampettu, Yue Yang, Ajay Patel, Joseph Chee Chang, Andrew Head.
CHI 2026 (in preparation).

Attribution Gradients: Incrementally Unfolding Citations for Critical Examination of Attributed AI Answers
Hita Kambhampettu, Alyssa Hwang, Philippe Laban, Andrew Head. *UIST 2026 (to appear).*

JumpStarter: Human-AI Planning with Task-Structured Context Curation
Xuanming Zhang, Sitong Wang, Jenny Ma, Alyssa Hwang, Zhou Yu, Lydia B Chilton.
EMNLP 2026 (under review).

FanOutQA: A Multi-Hop, Multi-Document Question Answering Benchmark for Large Language Models
Andrew Zhu, Alyssa Hwang, Liam Dugan, Chris Callison-Burch. *ACL 2024.*

RAID: A Shared Benchmark for Robust Evaluation of Machine-Generated Text Detectors
Liam Dugan, Alyssa Hwang, Filip Trhlik, Josh Magnus Ludan, Andrew Zhu, Hainiu Xu, Daphne Ippolito, Chris Callison-Burch. *ACL 2024.*

Ivie: Lightweight Anchored Explanations of Just-Generated Code
Litao Yan, Alyssa Hwang, Zhiyuan Wu, Andrew Head. *CHI 2024.*

NewsQs: Multi-Source Question Generation for the Inquiring Mind
Alyssa Hwang, Kalpit Dixit, Miguel Ballesteros, Yassine Benajiba, Vittorio Castelli, Markus Dreyer, Mohit Bansal, Kathleen McKeown. *preprint 2024.*

Kani: A Lightweight and Highly Hackable Framework for Building Language Model Applications
Andrew Zhu, Liam Dugan, Alyssa Hwang, Chris Callison-Burch. *EMNLP NLP-OSS Workshop 2023.*

Grounded Intuition of GPT-Vision's Abilities with Scientific Images
Alyssa Hwang, Andrew Head, Chris Callison-Burch. *preprint 2023.*

Rewriting the Script: Adapting Text Instructions for Voice Interaction
Alyssa Hwang, Natasha Oza, Chris Callison-Burch, Andrew Head. *DIS 2023.*

Large Language Models as Sous Chefs: Revising Recipes with GPT-3
Alyssa Hwang*, Bryan Li*, Zhaoyi Hou*, Dan Roth. *preprint 2023.*

** denotes equal contribution*

MEDIA COVERAGE

Featured in the *Tapping In* podcast, *Kent Place Magazine*, *Penn Today*, *Associated Press*, *TechCrunch*, and other outlets for human-AI interaction research.